

MU-HUA (MAURICE) WANG

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RESEARCH INTEREST

Physical assistive robotics for long-term care. I want to build robots that carry out contact-rich care tasks on a bedridden human body, starting with diaper changing and the cleaning that goes with it. The task demands three things at once: perceiving a body that will not cooperate, knowing enough about that particular person to move them without harm, and handling deformable material safely against skin. My work so far covers the first two and has begun the third. I built and clinically validated a markerless human motion capture system. I have studied, in musculoskeletal simulation, what a robot must know about a person before it moves their limb and what a learned model of a body is worth to a learner. I am now building a simulation platform for robot-assisted diaper changing. I am looking for a lab where these can be assembled into a machine that does the task.

EDUCATION

National Yang Ming Chiao Tung University (NYCU) — Taipei, Taiwan

M.S., Biomedical Engineering — entered Feb 2025; on approved leave of absence since May 2025 (enrollment retained)

Took leave to move from measuring human movement to building systems that act on it. Taught full time through the leave while starting the robot-learning work below, and now research full time.

B.S., Biomedical Engineering — Sep 2020 – Jun 2024

- GPA 3.72 / 4.30 · Final two years 4.07 / 4.30
- **IELTS 8.0 overall** — Reading 9.0, Listening 8.5, Speaking 7.5, Writing 7.0
- Coursework: Deep Learning for Biomedical Data Analysis; Neural Engineering; Brain and Physiological Signal Analysis; Smart Medical Device Mechanical Design; Medical Device Regulatory Affairs

RESEARCH EXPERIENCE

Project pages, videos and full manuscripts for everything below: maurice1128.github.io

I. Care-Robot Simulation Platform — Independent

2026 – present · in progress

Measuring Whether a Care Robot Hurts the Patient: a simulation platform for robot-assisted diaper changing and perineal cleaning

Existing caregiving simulators ask whether the robot can finish the task. This platform asks what the robot does to the body while trying.

- **Three axes, scored on every run:** is the garment on, is the perineum cleaned, is the patient injured.
- **The patient.** An SMPL-X body driven by MuJoCo, with pelvis, sacrum and femur meshes registered under the skin (hip-centre residual 10.4 mm). Every care position (supine, side-lying, frog-leg) is settled physically and must pass gates before it is used.
- **The tissue model.** A soft-tissue model of the sacral region with bone on the deep side. Its solver reproduces closed-form compression tests to 1.8×10^{-5} and its strain fields are converged across three mesh densities.
- **Tissue load.** After 285 simulated minutes supine, changing the mattress from rigid to 30 kPa foam lowers skin pressure by about 15% but shrinks the volume of deep tissue above the 285-minute strain level from 69 to 4 cm³. Interface pressure alone does not tell you about deep tissue.
- **Cleaning.** Of 798 legal robot bases, 19 reach all four targets, with peak contact force at most 2.8 N and no skin penetration. With a compliant cloth pad at 13–15 mmHg, 0.93 of the main perineal region is wiped at least three times under a criterion fixed before the run.
- **Garment.** A brief is closed around the settled body: front panel pulled up with stretch 1.24×, all four side tabs fastened.
- **Scope.** Simulation only, on a statistical body model; results are model-internal and not validated against measurements on people. Ongoing. Video and figures on the [project page](#).

II. Robot Learning and Embodied AI — Independent Research

Oct 2025 – present · Sole author on five manuscripts, four submitted

A self-directed programme, without an advisor or a lab, on one question: what must a robot know about a body, its own or the one it is touching, and what does that knowledge buy? Study design, experiments, statistics and verification are my own. Each result is tested against controls built to break it, and what it did not show is reported as a finding.

Gentle by Knowing: How Much Must a Care Robot Know About a Person to Lift Their Arm Safely? — *submitted, Machine Learning (Springer)*

- Robots that move a person's limb are protected today mainly by limits on contact force. I made the robot's knowledge of the person the independent variable. In a MyoSuite musculoskeletal simulation a robot supports the forearm from below and raises it 5 cm, for simulated care recipients whose shoulder or elbow is restricted; controllers differ only in what they know about the person and whether they may decline.
- No force-threshold guard reduced harmful outcomes by significantly more than the 5 percentage points fixed in advance. The guard's best setting cut them by 4–7 points at the cost of 40–51% of safe completions, because 95–100% of harmful outcomes began during the first attempt.
- A decision policy learned offline in simulation reduced harmful outcomes from 110 to 25 of 300 simulated persons at a similar number of safe completions (174 vs 177), using a care record with 20° error. Knowledge helped mainly by telling the robot whom not to lift.
- Reported against my own result: everything is simulated and harm is a mechanical exposure proxy; non-inferiority on safe completions was not established; and measured against perfect knowledge the policy both over- and under-refuses (47 of the 63 safe completions it missed were unnecessary refusals).

What a Motor Prior Is Worth: A Body Model and a Teacher Compared to Convergence and Through Withdrawal — *submitted, IEEE TCDS*

- The guidance hypothesis holds that guidance breeds dependence: the longer a learner is guided, the larger its fall when guidance is withdrawn. I tested it, and the same question for a model of the body, under one SAC learner on six muscle-driven bodies (two from MyoSuite, four planar arms of one to four joints), with at least twelve matched seeds per condition, an adversarial control for each prior, and training to a checked plateau.
- Replacing a withdrawn teacher by a frozen snapshot of the student keeps the loss term but adds no information, which separates what withdrawal removes: the teacher's share of the immediate drop falls with how long it was attached.
- Once training has converged, a learner that lost its teacher is never detectably worse than one never guided, at any attachment duration on either MyoSuite body. A withdrawn body model leaves the finger learner 15 mm short of one that kept it and 25 mm ahead of one that never had it.
- The babble-learned body model beats model-free learning on four of six bodies and a random model on all six.
- Reported against my own result: single-task reach in simulation; the body-model condition is handed the analytic reward at imagined states; the finger's first teacher was close to a no-op; and the contrasts were not pre-registered.

Modularity Makes Dexterous Hands Learn Faster, Not Better — *submitted, IEEE RA-L*

- Modular motion models promise a hand that learns faster and reaches further. I tested both promises on a simulated Shadow Hand, using each motion prior as the action space of a PPO policy, with the main comparisons fixed before the runs.
- The speed promise holds and the reach promise does not. At 1.5 M frames a keyframe interface with a learned in-betweeners scores 0.59 against a continuous latent's 0.22 on skill sequences never seen in training (17 of 20 seeds); by 5 M frames the latent has caught up (0.49 against 0.44). The same keyframes joined by a straight line never learned to release (0 of 10 seeds).
- Separately trained skills do no better than one flat policy trained on the same demonstrations (45 against 61 of 400 test episodes). The one gain, choosing the best policy for each stage (97 against 61), needs no separate training.
- Reported against my own result: one simulated hand, and the Adroit comparisons use one seed per task. The four conditions under which modularity would pay are offered as hypotheses.

A Hand-Motion Dataset Can Test Compositional Generalisation Only When Its Labels Describe the Scored Motion — *submitted, TMLR*

- Holding out a combination of labels is the usual test of whether a motion model generalises to unseen compositions. On recorded motion the split is made on labels, and whether holding out a label also holds out the motion had not been examined.

- Measured a paired penalty on four public hand-motion datasets and calibrated it against a synthetic zero-truth control (+2.6% of the naive error), planted interactions of known size, and permuted label grids on three of the datasets (+3.7%, +1.4%, −3.4%). The floor of the measurement depends on the dataset.
- On OakInk-Image, joining clips so that each correct label described only the first half of a trajectory, at fixed training volume, reduced the penalty from +17.0% to +3.9%, not distinguishable from the control. A penalty on TACO, written down before its data were opened, was found (+10.6%).
- Reported against my own result: per-seed estimates of the TACO penalty ranged from −3.8% to +27.4%; the error is an autoencoding error, so the result is about passing an unseen composition through a bottleneck, not about generating one; and one prior architecture was used.

Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer — draft

- An algebraic identity rules out a whole family of "reorganise the code" proposals before any experiment runs; the two conditions it needs are then measured. A re-encoding that preserves decodability by construction still costs +0.875 mm, a counterexample to decodability as a proxy for representation quality.

III. RGB Multi-View 3D Markerless Motion Capture — NYCU

Research Assistant, Oct 2023 – Sep 2024 · Team Lead, Oct 2024 – Sep 2025 Advisor: Prof. Hui-Ting Shih

Real-time whole-body pose sensing for clinical use — the perception layer a robot working over a bed has to have. I wrote the pipeline, ran the validation experiments, and collected the clinical data myself.

- **Built the real-time pipeline end to end.** GPU acceleration and CPU parallelism across capture, inference, triangulation and inverse kinematics; DLT triangulation with occlusion handling; and **ROWV**, an outlier-rejection algorithm for small-sample settings, faster and more accurate than KNN and RANSAC on this problem.
- **Ran the validation experiments myself.** VICON comparison over 21 participants, plus a study over 104 walking trials from 11 participants on a nine-camera ring showing that per-joint verdicts between triangulation variants depend on the joint-centre offset they contain.
- **Ran the field data collection.** 200+ stroke and knee-pain patients (Dr. Ho, Vietnam); 40+ stroke patients with exoskeleton (Kaohsiung Chang Gung Memorial Hospital); 20+ elderly participants (Kaohsiung daycare centre); real-time feedback trials with lower-limb amputees (NYCU Physical Therapy and Assistive Technology). Hospital wards, clinics and daycare centres, not a laboratory.
- **Asked how much sensing error the downstream decision can absorb** — forward-dynamic simulation with a Geyer–Herr reflex controller over 66 design cells, showing identification of a weak muscle group falls from 0.845 to 0.549 as joint-angle error grows $0^\circ \rightarrow 12^\circ$.
- **Shipped the whole product**, not only the algorithm: clinician-facing GUI (PyQt6), SQL and cloud pipeline with ICD-10 coding for longitudinal and cross-sectional search, and an Unreal Engine rehabilitation game.
- **Led a 5-person R&D team** in the second year and drove the system into clinical deployment: 4 conference presentations and 6 media features (CNA, PTS News, Yahoo! News Taiwan, iThome, OCAC, NYCU).
- **Ran the collaborations**, across institutions that do not normally open up to a student: 2 signed MOUs (Taipei Veterans General Hospital; Taipei City Hospital), Kaohsiung Chang Gung Memorial Hospital, the Institute for Information Industry (cloud patient database with ICD-10 coding), an international site in Vietnam, a community daycare centre and a private clinic in Kaohsiung, a sports-performance company, and NYCU Physical Therapy and Assistive Technology.

IV. Circular-Geometry Representation of Behavioural Trajectories — Taoyuan General Hospital

Jan 2023 – present · Advisor: Dr. Tso-Jeng Wang

A methods problem disguised as a clinical one: the discriminating feature of a dose trajectory is its *shape*, but shape is confounded with magnitude and lives on a space that Euclidean clustering gets wrong.

- **Built the representation.** Normalised each daily consumed-dose series to its own value seven days earlier, took the extreme three-day moving-window slopes on each side of the test, and mapped the (post, pre) slope pair to a polar angle $\theta = \text{atan2}(\text{post}, \text{pre})$. Shape becomes comparable independent of absolute magnitude, and the feature space becomes a torus.
- **Clustered on the manifold, not in the plane.** Fitted a bivariate (product) von Mises mixture by EM on the circle, because Euclidean clustering does not respect wrap-around at $0^\circ/360^\circ$. For the stable-dose branch, reduced 114 variables by PCA and cut a circular kernel-density estimate at its valleys. Nine patterns emerged, spanning positive/negative occurrence ratios 1.64 to 0.50.
- **Attacked my own result.** Outcome-blind re-clustering, 500 bootstrap refits per branch, within-patient conditional logistic regression, restriction to windows containing no interpolated day, and reporting across $K = 4\text{--}8$ because K was an investigator choice and no criterion selected it.

- **Reported what did not survive.** Post-test escalation magnitude (OR 1.36 → 1.06 within patients) is not claimed; the stable-dose branch is labelled exploratory; the association did not replicate across a temporal split; and the trajectory adds essentially nothing predictive over attendance. Pre-test escalation did survive, and strengthened (1.15 → 1.26 per SD, $p = 3 \times 10^{-7}$).
- Scale: 9,900 analysable urine tests from 552 patients over 2018–2022, from a 20,108-record extract; earlier phase covered 1,000+ patients from the Taiwan National Health Insurance Database.

V. EEG and EMG Sleep–Epilepsy Semi-Automated Analysis — NYCU

Oct 2022 – Oct 2023 · Advisor: Prof. You-Yin Chen

- Built a signal pipeline processing 24-hour recordings in 3 minutes (detrending, downsampling, filter design, time–frequency conversion, outlier detection) with automatic WAKE / NREM / REM staging from band-specific energy features, plus a manual-scoring GUI for cross-labeler consistency.
- Designed a surgical modification that raised EMG signal-to-noise ratio at the source. Collaboration with Hualien Tzu Chi Hospital and Taipei Medical University.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed conference papers

1. **Wang, M.-H.**, Tsai, K.-T., Chen, B.-H., Huang, L.-W., Tsai, C.-H., Hsu, C.-S., & Chen, Y.-Y. *Markerless Gait Analysis for Lower Back Pain: A Cost-Effective Solution to Assess Pelvis–Trunk Coordination*. **SEMBA 2025**. (first author)
2. Huang, L.-W., **Wang, M.-H.**, Chen, B.-H., Su, K.-H., Tsai, C.-H., Chen, Y.-Y., & Shih, H.-T. *A Real-Time Algorithm for Detecting Gait Events in Lower-Limb Prosthesis Users Using a Markerless Motion Capture System*. **ASB 2025**.
3. Huang, A.-Y., **Wang, M.-H.**, Huang, L.-W., Chen, B.-H., Wang, H.-H., Kuo, C.-H., Lu, S.-W., & Lo, Y.-C. *Spatiotemporal Cortical-Synergy Connectivity in the Gait Cycle*. **SEMBA 2025**.
4. Chang, C.-W., **Wang, M.-H.**, Lin, Y.-C., Wang, C.-Y., Li, S.-J., Chen, T.-C., Liang, Y.-W., Chen, C.-T., Chen, Y.-Y., & Lin, S.-H. *Investigating the Effect of Central Thalamic Deep Brain Stimulation on Sleep in an Alzheimer's Disease Model*. **ISMIRM 2024**.
5. Chen, B.-H., Tsai, C.-H., Huang, L.-W., **Wang, M.-H.**, Hsu, C.-S., & Chen, Y.-Y. *Robust Self-Occlusion Mitigation for Enhanced Accuracy in Multi-View 3D Markerless Motion Capture Systems*. **ICMST 2024**.

Submitted

- S1. **Wang, M.-H.** (sole author). *Gentle by Knowing: How Much Must a Care Robot Know About a Person to Lift Their Arm Safely?* **Machine Learning** (Springer), submitted 2026.
- S2. **Wang, M.-H.** (sole author). *What a Motor Prior Is Worth: A Body Model and a Teacher Compared to Convergence and Through Withdrawal*. **IEEE Transactions on Cognitive and Developmental Systems**, submitted 2026. Merges and supersedes the earlier draft *Two Routes to a Motor Prior*.
- S3. **Wang, M.-H.** (sole author). *Modularity Makes Dexterous Hands Learn Faster, Not Better*. **IEEE Robotics and Automation Letters**, submitted 2026.
- S4. **Wang, M.-H.** (sole author). *A Hand-Motion Dataset Can Test Compositional Generalisation Only When Its Labels Describe the Scored Motion*. **Transactions on Machine Learning Research**, submitted 2026. Supersedes the earlier draft *Which Hand-Motion Composition Axes Are Worth Measuring*.

Manuscript drafts — on the website

Robot learning

- D1. **Wang, M.-H.** (sole author). *Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer*.

Human motion sensing and clinical biomechanics

- D2. **Wang, M.-H.**, & Shih, H.-T. *Identification of a Weak Muscle Group Tolerates More Joint-Angle Error than Kinematic Reliability Conventions Allow*.
- D3. **Wang, M.-H.**, & Shih, H.-T. *Joint-Centre Offset Dominates 3D Position Error and Can Reverse Comparisons Between Markerless Triangulation Methods*.
- D4. **Wang, M.-H.**, & Shih, H.-T. *Distinguishing Graded Plantarflexor Hyperreflexia from Graded Dorsiflexor Weakness in Simulated Post-Stroke Gait: A Search over 81 Sagittal Kinematic Variables*.

Behavioural data science

- D5. **Wang, M.-H.**, & Wang, T.-J. *Consumed Methadone Dose Patterns Around Randomly Scheduled Urine Testing: Pre-Test Escalation and Higher-Risk Trajectories in Long-Retained Maintenance Patients*.

TEACHING

Physical Science Teacher (Grade 8) — Tamkang High School, New Taipei · Feb 2026 – Aug 2026

Taught the full Grade-8 physical science curriculum to two of the grade's eight classes.

- **Class 8 finished 2nd of eight classes on the third term exam, up from 3rd**, with its mean margin over the school average widening from +5.1 to +11.8 points across three term exams.
- **The gain came from the bottom of the distribution, not the top.** By the third exam, Class 8's bottom-quartile cutoff (78) had reached the school-wide median, and its bottom-12% cutoff stood 28 points above the school's — up from a 14-point gap on the first exam.
- Class 8 held the second-narrowest score spread in the grade (SD 15.07 against a school-wide 21.86): the class improved by pulling up students who were behind, not by stretching students already ahead.
- **Designed the mechanism that produced it.** The class earned its reward only if *every* student scored 100 — a 10-minute paper, then open discussion, then a fresh paper, repeated until the whole class cleared it. The collective condition put the strongest students to work teaching the weakest, which is where the measured gain came from.
- **Ran self-initiated weekly remedial sessions** outside class hours across three grades, targeted at the students who were behind — not assigned by the school.
- Ran hands-on chemistry and physics activities outside the syllabus, and class events to build cohesion.

HONORS AND AWARDS

2nd Place — Youth AI Startup Collaborative Hackathon, Ministry of Digital Affairs · Fall 2023 · Taipei

Led a team building the business model and commercialization path for a markerless visible-light motion capture system. Prize US\$1,000.

1st Place — Competition for Innovative Proposals in Elderly Leisure, Health and Technology Practices, National Chi Nan University · Fall 2023 · Nantou

Proposed a fall-detection and emotional-gait-recognition model for elderly care. Prize US\$340.

TECHNICAL SKILLS

Languages	Python (advanced), MATLAB (advanced), C/C++ (intermediate)
ML / RL	PyTorch, TensorFlow; SAC, PPO and Dyna-style model-based RL; policy distillation; offline policy learning; world models; behaviour cloning; CNN/VGG; LSTM
Robotics & simulation	MuJoCo; MyoSuite; Shadow Hand; Adroit; SMPL-X; musculoskeletal forward dynamics; Geyer–Herr reflex control; inverse kinematics; DLT triangulation; Unreal Engine
High-performance computing	CuPy, Numba, multiprocessing, GPU acceleration, CPU affinity and priority tuning
Statistics & experiment design	matched-seed paired designs; predictions fixed before the runs; zero-truth, planted and permuted-label controls; adversarial controls; non-inferiority and equivalence margins; von Mises / directional statistics; bootstrap reproducibility; conditional logistic regression
Systems	PyQt6, SQL, cloud data pipelines, ICD-10 clinical coding

LEADERSHIP AND ACTIVITIES

- **Leader, Rural Service Team** · Fall 2024 – Fall 2025 — led two 3-day service camps in Tamsui and Fenglin combining educational workshops with community service. **Programme lead, Beitou church service team** (2026).
- **Music Director, church worship team** · present — and **Founder & President, ToGather Music (YouTube)** · Fall 2021 — a team of 3 producers and 5 vocalists, a self-built recording studio, 13 original songs published. Five years of continuous responsibility for a performing team.
- **Long-term mentor** to a group of four students spanning primary to senior-high school.
- **Chief Organizer, Biomedical Engineering & Nursing Student Camp** · Fall 2022 — 50 students across two disciplines. **Graduation Representative** · Fall 2024.